Reviewer 1

Novelty: 6 (reasonably novel - there are some notable differences from existing ideas and probably enough to turn into a new paper)
Rationale: I haven't seen (and couldn't find) any prior work which exactly has the same idea as in this proposal. The proposed idea is definitely related to using consistency among multiple solutions to estimate uncertainty (e.g. https://arxiv.org/abs/2405.18711 does this across solutions decoded from different layers) but I have not seen the idea of constructing resonance graph and using graph properties to estimate uncertainty.

Feasibility: 8 (Highly Feasible: Straightforward to implement the idea and run all the experiments.)
Rationale: The proposed method, SRUQ, should be pretty easy to implement given that LLM API access is abundant. SRUQ involves multiple steps all of which can be done through prompting via API — getting multiple solutions, prompting LLMs to get a consistency score between each pair of solutions etc. The parts which cannot be implemented through API are the baselines e.g. Monte Carlo dropout, and would require GPUs. To do a fair comparison to the baselines, I imagine SRUQ will also have to be done on open models which could also require GPUs.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)
Rationale: Although the proposal includes some baselines that should be compared to, it does not mention some methods which seem to do quite well with LLMs (especially getting better with scale) – e.g. methods like P(True) (https://arxiv.org/abs/2207.05221) or verbalized confidence (https://arxiv.org/abs/2305.14975). It's not clear/obvious to me that the proposed method should do better than these baselines.

Excitement: 6 (Learning positive: exciting enough to be accepted at a major AI conference, but still has some weaknesses or somewhat incremental)
Rationale: While the method is novel and feasible, I'm not too excited by it since some of the other existing methods out there mentioned above (like https://arxiv.org/abs/2207.05221, https://arxiv.org/abs/2305.14975) are much simpler and work quite well. Compared to that SRUQ is more complex, and hence maybe has less chance of being very impactful (unless it works really better).

Overall Score: 6 (Marginally above the acceptance threshold of major AI conferences)
Rationale: The above accept score is assuming the idea does work better than the baselines on some category of tasks. Overall, given that the idea is novel, the proposal includes comparison to other baselines as well analysis & ablations, I think that could be enough to get accepted into an AI conference.

Confidence: 4 (You are confident but not absolutely certain that the evaluation is correct)

Reviewer 2

Novelty: 6 (reasonably novel - there are some notable differences from existing ideas and probably enough to turn into a new paper)
Rationale: The proposed approach shares some similar ideas with self-consistency (which suggests the consistency of sampled LLMs outputs is relatively well calibrated). But the approach is more generalized and fine-grained than existing work if the approach uses more advanced 'mutual support evaluation' beyond simply comparing the final answers.

Feasibility: 5 (Moderately feasible: It can probably be executed within the given time frame but would require careful planning, efficient use of APIs or some advanced computational strategies to overcome the limited GPU resources, and would require some modifications to the original proposal to make it work.)
Rationale: There lacks some important details in terms of the cross-evaluation part. How is the mutual support evaluated (by prompting or some other methods?). This part is crucial for implementing the whole pipeline of this approach.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)
Rationale: I think it has some chances to beat the proposed baselines. If the cross-evaluation part is properly executed. Again, the success of this proposal is highly dependent on that part.

Excitement: 6 (Learning positive: exciting enough to be accepted at a major AI conference, but still has some weaknesses or somewhat incremental)
Rationale: If this idea actually works, at least it tells something new about how to use multiple samples to provide better confidence estimation than simple consistency. But the idea itself is still somewhat incremental given the existence of current consistency-based calibrators.

Overall Score: 6 (Marginally above the acceptance threshold of major AI conferences)
Rationale: Overall there are some incremental contributions, but not too exciting. The algorithm itself can be neat. I think it can be worth a borderline acceptance.

Confidence: 4 (You are confident but not absolutely certain that the evaluation is correct)

Reviewer 3

Novelty: 6 (reasonably novel - there are some notable differences from existing ideas and probably enough to turn into a new paper)
Rationale: I think the idea is reasonable and indeed identifies some limitations of current works on uncertainty estimation. However, the consistency between reasoning paths is somehow similar to self-consistency reasoning from Google and SelfCheckGPT.

Feasibility: 7
Rationale: I think it could be easy to implement and quickly be tried by PhD students or even undergrads. Also, in the test case example, the setting is straightforward and well-defined.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)
Rationale: Based on my experience, the consistency-based methods, although not fully theoretically grounded, can work pretty well in current uncertainty estimation questions. I believe working this on the reasoning path level could also work to some extent.

Excitement: 6 (Learning positive: exciting enough to be accepted at a major AI conference, but still has some weaknesses or somewhat incremental)
Rationale: Overall, this idea identified a good research question, although the method might not be very exciting to me.

Overall Score: 6 (Marginally above the acceptance threshold of major AI conferences)
Rationale: The novelty and the actual application of this method in the area is limited, but could be an inspiring idea.

Confidence: 4 (You are confident but not absolutely certain that the evaluation is correct)

R Example Idea: Translation with LLMs through Prompting with Long-Form Context

Translation with LLMs through Prompting with Long-Form Context (Part 1)

1. Problem Statement: Stable generation of text in low-resource languages is an unsolved issue in large language models.

2. Motivation: While LLMs can often produce surprisingly good translations despite not being explicitly trained for this task, this does not hold for lower-resource languages. LLMs are both more likely to generate off-target text (text in another language than intended) when prompted to translate to a lower-resource language, and show increased instability in translation quality across prompt templates in lower-resource languages.

3. Proposed Method: Our proposed method investigates the use of long-form templates to improve generated translation quality and reduce off-target translations in lower-resource languages. We propose to provide additional prompt context by translating multi-sentence input, with additional views of the target language with the langid template provided as context. We do so in multiple stages:

1. **Querying the language model** to first generate a paragraph containing the source sentence to be translated.
2. **Prepending monolingual text in the target language**, with langid: tags, above the translation prompt.
3. **Presenting both these additional sources of content**, prompting the LLM for a translation.

4. Step-by-Step Experiment Plan:

1. **Choose datasets:** Evaluate on the FLORES-200 datasets, which allow for wide language coverage on the Wikipedia domain, as well as the WMT-21 test sets for news and law/medical domain.
2. **Choose languages:** Opt for English-centric translation with:
   - 5 high-resource languages with different scripts (French, German, Russian, Chinese, Japanese)
   - 5 mid-resource languages (Farsi, Vietnamese, Arabic, Korean, Hebrew)
   - 5 low-resource languages with considerably lower likelihood of incidental bilingualism (Gujarati, Thai, Tajik, Sindhi, Pashto)
3. **Choose models:** Include the API-based GPT-3.5 (Text-Davinci-003) and GPT-4 model from OpenAI and Gemini from Google, as well as the open-weight LLaMA-3, Gemma, and Aya models which enable additional analysis.
4. **Gather translation results:** Systematically compare standard MT prompt templates to our proposed method across different models and language pairs. Additionally ablate the steps of the new method (removing langid templates; replacing langid templates with endonymic langid tags; provide only the generated paragraph; only the monolingual content).
5. **Perform analysis:** Evaluate whether the new method improves the performance of LLMs in these tasks as compared to the baselines using multiple standard automatic metrics for MT (chrF, COMET, BLEU) and token-level LID to measure off-target translations. Assess which component(s) are necessary for this improvement and whether or not there are changes across language pair direction and language resource levels or scripts.

Translation with LLMs through Prompting with Long-Form Context (Part 2)

5. Test Case Example:

- **Baseline Prompt Template:**

```
[English] This is an example.
[Spanish]
```

- **Baseline Prompt Expected Output:**

```
[Spanish] Este es un ejemplo.
```

- **Proposed Prompt Input (step 1):**

```
Write a paragraph containing the following sentence:
[English] This is an example.
```

- **Expected model output (step 1):**

```
This is an example link. Italics and Bold: use asterisks and underscores
to indicate spans of emphasis. Use one asterisk (or underscore) for
italics and two asterisks (or two underscores) for bold. For example...
```

- **Proposed Prompt Input (step 2):**

```
[Spanish] Computadoras simples son lo suficientemente pequeñas
para residir en los dispositivos móviles.
[Spanish] Las computadoras portátiles, tales come tabletas,
netbooks, notebooks, ultrabooks, pueden ser alimentadas por
pequeñas baterías.
[Spanish] Las computadoras personales en sus diversas formas son
iconos de la llamada era de la información y son lo que la mayoría
de la gente considera como «computadora».

Translate this paragraph from English to Spanish:
[English] This is an example link. Italics and Bold: use asterisks
and underscores to indicate spans of emphasis. Use one asterisk
(or underscore) for italics and two asterisks (or two underscores)
for bold. For example...
[Spanish]
```

- **Proposed Prompt Expected Output:**

```
[Spanish] Este es un enlace de ejemplo. Cursiva y negrita: utilice
asteriscos y guiones bajos para indicar intervalos de énfasis.
Utilice un asterisco (o guión bajo) para cursiva y dos asteriscos
(o dos guiones bajos) para negrita. Por ejemplo...
```